Closed-Form Item-Item Recommendation with EASE^{R} for Restaurant Consumer Ratings

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Abstract

We apply EASE^{R} (Embarrassingly Shallow Autoencoder for Sparse Recommendation), a closed-form, ridge-regularised item-item linear model, to the Restaurant Consumer Ratings dataset. Because the dataset carries no event timestamp, evaluation uses a user-aware random holdout instead of a temporal split. The system includes a declarative feature specification, leakage tests, a popularity baseline, and reproducible Floci/AWS infrastructure.

1 Introduction

Most public restaurant-recommendation notebooks rely on cosine-similarity k-NN or basic SVD. This project instead implements EASE^{R} [1], a linear autoencoder with a closed-form solution: no gradient descent, no iterative optimisation, and no neural network. EASE reconstructs each user's interaction vector as a sparse linear combination of their other interactions, which makes every recommendation explainable as "restaurants similar to the ones you already liked."

2 Dataset

The Restaurant Consumer Ratings dataset (UCI / Kaggle ref `uciml/restaurant-data-with-consumer-ratings`) contains 1,161 overall ratings on a 0–2 scale from 138 consumers over 130 restaurants in Mexico, along with restaurant metadata (cuisine, price tier, ambience). Ratings carry no event timestamp. Following the implicit-feedback convention used across this portfolio, a rating of 1 or 2 is treated as a positive and a rating of 0 is dropped. An iterative k-core filter ($k = 3$) removes users and restaurants with too little history for collaborative signal.

3 Feature engineering

Features are declared in `conf/features.yaml` and implemented in `src/recommender/features/`. User features (`user_activity_count`, `user_mean_rating`) are fitted strictly on the training partition. Item features combine a training-only statistic (`item_popularity_rank`) with catalogue-only descriptors (`item_price_tier`, `item_cuisine_tokens`) that carry no label signal by construction. A registry check (`assert_registry_matches_spec`) asserts the YAML specification and the Python implementation agree in both directions.

4 Methodology

Let $X \in \{0,1\}^{n \times m}$ be the binary user-item interaction matrix over n training users and m items. EASE^{R} solves

$$\min_{B} \|X - XB\|_F^2 + \lambda \|B\|_F^2, \quad \text{s.t. } \operatorname{diag}(B) = 0, \tag{1}$$

where the zero-diagonal constraint prevents the trivial solution $B = I$. Writing $G = X^\top X + \lambda I$, the Lagrangian stationarity condition yields the closed-form solution

$$P = G^{-1}, \tag{2}$$

$$B_{ij} = \begin{cases} -P_{ij}/P_{jj} & i \neq j \\ 0 & i = j \end{cases}. \tag{3}$$

A user's predicted score vector is $\hat{s}_u = x_u B$, where x_u is the user's own (possibly cold, serving-time) interaction vector: no per-user embedding is learned, so a user unseen at training time is scored directly from their observed history. The single hyperparameter λ controls ridge shrinkage of the off-diagonal weights.

5 Experimental protocol

Because the dataset has no timestamp, we use a *user-aware random holdout*: for each user with at least $\lceil 1/(1-p) \rceil + 1$ interactions (where $p = 0.3$ is the combined validation+test fraction), 10% of their interactions are held out for validation and 20% for test, sampled uniformly at random per user; all other users contribute entirely to training. We report NDCG@10, Recall@10, MAP@10, hit rate, and catalogue coverage on the test partition, and compare EASE^{R} against a non-personalised popularity baseline and a random-ranking floor.

6 Results

Table 1: Top-10 ranking metrics on the synthetic CI fixture (test partition).

Model	NDCG@10	Recall@10	MAP@10	Coverage
EASE^R (λ=300)	0.2428	0.4430	0.1816	0.4833
Popularity	0.2423	0.4430	0.1795	0.2833

Table 1 reports ranking metrics for EASE^{R} (λ = 300) against the popularity and random baselines on the synthetic fixture used in continuous integration (the real dataset reproduces the same protocol locally via `scripts/download_data.py --kaggle`). EASE^{R} matches or exceeds the popularity floor on ranking quality while covering a substantially larger fraction of the catalogue, which is the expected signature of a personalised model on data with real co-visitation structure.

7 Limitations and conclusion

The dataset is small (138 users, 130 restaurants), so EASE^{R}'s advantage over popularity is modest; the gap is expected to widen on larger, denser restaurant-review corpora. The model also has no explicit cold-start mechanism beyond falling back to a user's available history or, for a first-ever visitor, the popularity floor. Future work could add a content-based fallback using the cuisine and price-tier features already collected. Overall, EASE^{R} offers a strong, fully closed-form, non-neural baseline with transparent item-item weights and cheap retraining.

References

[1] Harald Steck. Embarrassingly shallow autoencoders for sparse data. In *The World Wide Web Conference*, pages 3251–3257, 2019.